

# 3-DOF Vehicle Dynamics Implementation: From 1-DOF to Full Planar Motion

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## Abstract

This document describes the implementation of a full 3-DOF (longitudinal, lateral, yaw) vehicle dynamics model, extending the existing 1-DOF Dugoff-based tire model. We present the rigid body equations of motion, per-wheel kinematics accounting for vehicle rotation, coordinate frame transformations for steered wheels, and tire force sign conventions. The architecture integrates with a CAN-based control loop and explicit gear selector for forward/reverse operation.

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# 1 Introduction

## 1.1 From 1-DOF to 3-DOF

The base vehicle model (documented in the companion Dugoff tire model document) provides:

- 1-DOF longitudinal dynamics with Dugoff tire forces
- Wheel rotational dynamics and slip ratio computation
- Normal load distribution with longitudinal load transfer
- Ackermann steering geometry for front wheels

This document extends the model to **full planar motion (3-DOF)**:

$$\text{State: } \mathbf{x} = [v_x, v_y, \dot{\psi}, x, y, \psi]^\top \quad (1)$$

where:

- $v_x$  = longitudinal velocity (body frame)
- $v_y$  = lateral velocity (body frame)
- $\dot{\psi}$  = yaw rate
- $x, y$  = position (global frame)
- $\psi$  = heading angle (yaw)

## 1.2 Motivation

1-DOF models cannot capture:

- **Lateral dynamics:** Cornering forces, sideslip, understeer/oversteer
- **Yaw dynamics:** Vehicle rotation from steering and lateral tire forces
- **Combined slip:** Interaction between longitudinal and lateral forces
- **Maneuver testing:** Slalom, lane change, emergency avoidance

The 3-DOF model enables realistic testing of:

- Path tracking controllers
- Stability control systems
- Emergency maneuvers (double lane change, moose test)
- Combined braking-steering scenarios

# 2 Rigid Body Dynamics (3-DOF)

## 2.1 Equations of Motion

The vehicle is modeled as a rigid body in the plane with three degrees of freedom. All velocities and accelerations are expressed in the **body-fixed frame** (origin at CG, x-axis forward, y-axis left).

### 2.1.1 Longitudinal Dynamics

$$m\dot{v}_x = F_{x,\text{total}} - F_{\text{drag}} - F_{\text{roll}} + mv_y\dot{\psi} \quad (2)$$

where:

$$F_{x,\text{total}} = F_{x,fl} + F_{x,fr} + F_{x,rl} + F_{x,rr} \quad (3)$$

$$F_{\text{drag}} = c_d v_x |v_x| \quad (4)$$

$$F_{\text{roll}} = c_r \cdot \text{sgn}(v_x) \quad (5)$$

**Critical Note:** The term  $mv_y\dot{\psi}$  is the **\*\*centrifugal force\*\*** arising from the rotating reference frame. It is often omitted in simplified models, leading to instability.

### 2.1.2 Lateral Dynamics

$$m\dot{v}_y = F_{y,\text{total}} - mv_x\dot{\psi} \quad (6)$$

where:

$$F_{y,\text{total}} = F_{y,fl} + F_{y,fr} + F_{y,rl} + F_{y,rr} \quad (7)$$

The term  $-mv_x\dot{\psi}$  is the **\*\*centripetal acceleration\*\*** in the body frame. This coupling between longitudinal and lateral dynamics is **\*\*essential for stability\*\***.

### 2.1.3 Yaw Dynamics

$$I_z\ddot{\psi} = M_{z,\text{total}} \quad (8)$$

where the yaw moment is computed from the cross product  $\mathbf{r}_i \times \mathbf{F}_i$  for each wheel:

$$M_{z,\text{total}} = \sum_{i=1}^4 (x_i F_{y,i} - y_i F_{x,i}) \quad (9)$$

with wheel positions:

$$\text{Front left: } \mathbf{r}_{fl} = [x_f, +w/2]^\top \quad (10)$$

$$\text{Front right: } \mathbf{r}_{fr} = [x_f, -w/2]^\top \quad (11)$$

$$\text{Rear left: } \mathbf{r}_{rl} = [-x_r, +w/2]^\top \quad (12)$$

$$\text{Rear right: } \mathbf{r}_{rr} = [-x_r, -w/2]^\top \quad (13)$$

where  $x_f$  is CG distance to front axle,  $x_r$  is CG distance to rear axle, and  $w$  is track width.

## 2.2 Position Integration

Vehicle position in the global frame is computed by transforming body velocities:

$$\dot{x} = v_x \cos \psi - v_y \sin \psi \quad (14)$$

$$\dot{y} = v_x \sin \psi + v_y \cos \psi \quad (15)$$

$$\dot{\psi} = \dot{\psi} \quad (16)$$

This is a standard 2D rotation matrix  $R(\psi)$  applied to  $[v_x, v_y]^\top$ .

### 3 Per-Wheel Kinematics

#### 3.1 Velocity at Wheel Center

Each wheel experiences a different velocity due to:

1. CG lateral velocity  $v_y$
2. Yaw rate  $\dot{\psi}$  causing tangential velocity at offset from CG

The velocity at wheel  $i$  in the **vehicle frame** is:

$$\mathbf{v}_i = \begin{bmatrix} v_x \\ v_y \end{bmatrix} + \begin{bmatrix} -\dot{\psi}y_i \\ \dot{\psi}x_i \end{bmatrix} = \begin{bmatrix} v_x - \dot{\psi}y_i \\ v_y + \dot{\psi}x_i \end{bmatrix} \quad (17)$$

For steered wheels (front), we must transform this to the **wheel frame** (aligned with wheel heading):

$$\begin{bmatrix} v_{x,w} \\ v_{y,w} \end{bmatrix} = \begin{bmatrix} \cos \delta & \sin \delta \\ -\sin \delta & \cos \delta \end{bmatrix} \begin{bmatrix} v_x - \dot{\psi}y_i \\ v_y + \dot{\psi}x_i \end{bmatrix} \quad (18)$$

where  $\delta$  is the wheel steer angle.

#### 3.2 Slip Angle Computation

The slip angle  $\alpha$  is the angle between the wheel heading and its velocity direction:

$$\alpha = \arctan \left( \frac{v_{y,w}}{|v_{x,w}|} \right) \quad (19)$$

Note: We use  $|v_{x,w}|$  (absolute value) to handle both forward and reverse motion correctly.

### 4 Tire Force Coordinate Transformations

#### 4.1 Rotation Matrix for Steered Wheels

The Dugoff tire model computes forces  $[F_x, F_y]^\top$  in the **wheel coordinate frame** (aligned with the wheel's heading). For steered wheels, these forces must be transformed to the **vehicle frame** before summing.

The standard 2D rotation matrix to express a wheel-frame vector in vehicle-frame coordinates is:

$$\begin{bmatrix} F_{x,veh} \\ F_{y,veh} \end{bmatrix} = \begin{bmatrix} \cos \delta & -\sin \delta \\ \sin \delta & \cos \delta \end{bmatrix} \begin{bmatrix} F_{x,wheel} \\ F_{y,wheel} \end{bmatrix} \quad (20)$$

Expanding:

$$F_{x,veh} = F_{x,wheel} \cos \delta - F_{y,wheel} \sin \delta \quad (21)$$

$$F_{y,veh} = F_{x,wheel} \sin \delta + F_{y,wheel} \cos \delta \quad (22)$$

**Physical interpretation:** When the wheel is steered left ( $\delta > 0$ ) and experiences a leftward force ( $F_{y,wheel} > 0$ ), the transformation correctly produces a leftward force component in the vehicle frame, contributing to the intended turn direction.

### 5 Tire Force Sign Conventions

#### 5.1 Fundamental Principle

Tire forces must **oppose slip** in all directions. This is the essence of friction.

## 5.2 Longitudinal Force

### 5.2.1 Slip Ratio Definition

$$\sigma_x = \frac{\omega R - V_x}{\max(|V_x|, |\omega R|, v_{\min})} \quad (23)$$

Note: Denominator uses  $|V_x|$  to prevent division by zero and handle reverse motion correctly.

### 5.2.2 Force Formula with Gear Direction

$$F_x = -g_{\text{dir}} \cdot C_x \sigma_x f(\lambda) \quad (24)$$

where  $g_{\text{dir}}$  is the **\*\*gear direction\*\*** from CAN command:

- Forward:  $g_{\text{dir}} = +1$
- Reverse:  $g_{\text{dir}} = -1$
- Neutral:  $g_{\text{dir}} = 0$

**Physical interpretation:**

- Forward gear,  $\sigma_x > 0$  (wheel spinning faster):  $F_x < 0$  (brakes vehicle)
- Forward gear,  $\sigma_x < 0$  (wheel locked):  $F_x > 0$  (resists locking, ABS effect)
- Reverse gear,  $\sigma_x > 0$ :  $F_x > 0$  (resists reverse motion)
- Neutral:  $F_x = 0$  (no longitudinal traction)

## 5.3 Lateral Force

$$F_y = -C_y \sigma_y f(\lambda) \quad (25)$$

where  $\sigma_y = V_y/|V_x|$  is the lateral slip ratio (small-angle approximation of slip angle).

Lateral force is **\*\*independent\*\*** of gear direction. It always opposes lateral slip regardless of forward/reverse motion. The negative sign ensures the force resists slip: if the wheel slips left ( $\sigma_y > 0$ ), the force pushes right ( $F_y < 0$ ).

## 6 Yaw Moment Calculation

### 6.1 Cross Product Formula

For a force  $\mathbf{F} = [F_x, F_y]^\top$  at position  $\mathbf{r} = [x, y]^\top$  relative to the CG, the yaw moment (z-component of  $\mathbf{r} \times \mathbf{F}$ ) is:

$$M_z = xF_y - yF_x \quad (26)$$

**Physical example:** A leftward force ( $F_y > 0$ ) at the front-left wheel with position ( $x_f > 0, y > 0$ ) creates:

$$M_z = x_f F_y - y \cdot 0 = x_f F_y > 0 \quad (\text{CCW yaw, left turn}) \quad (27)$$

This is correct: a left force on the left side of the vehicle creates a left (counterclockwise) turn.

## 7 Centripetal Coupling in Rotating Reference Frame

### 7.1 Equations in Body Frame

In the vehicle body frame (rotating with the vehicle), both longitudinal and lateral velocities are affected by rotation:

$$\dot{v}_x = \frac{F_x}{m} + v_y \dot{\psi} \quad (\text{centrifugal acceleration}) \quad (28)$$

$$\dot{v}_y = \frac{F_y}{m} - v_x \dot{\psi} \quad (\text{centripetal acceleration}) \quad (29)$$

The terms  $v_y \dot{\psi}$  and  $-v_x \dot{\psi}$  arise from the non-inertial (rotating) reference frame. They are **essential for stability** and must be included in the integration.

### 7.2 Physical Interpretation

When the vehicle turns (yaw rate  $\dot{\psi} \neq 0$ ):

- **Centrifugal effect** ( $v_y \dot{\psi}$ ): Lateral velocity creates an apparent longitudinal acceleration in the body frame
- **Centripetal effect** ( $-v_x \dot{\psi}$ ): Longitudinal velocity requires centripetal acceleration to maintain circular motion

These terms couple the longitudinal and lateral dynamics, making the 3-DOF model fundamentally different from independent 1-DOF models.

## 8 CAN-Based Control Architecture

### 8.1 Gear Selector Implementation

To properly handle forward/reverse motion, an explicit **gear selector** was implemented via CAN (matching real autonomous vehicle architecture).

#### 8.1.1 CAN Signal Definition

Frame: ACTUATOR\_CMD\_1 (0x100), 10 ms cycle

| Signal              | Start Bit | Length | Values      | Unit |
|---------------------|-----------|--------|-------------|------|
| system_enable       | 0         | 1      | 0/1         | bool |
| gear_position       | 1         | 2      | 0-3         | enum |
| mode                | 3         | 2      | 0-3         | enum |
| steer_cmd_deg       | 8         | 16     | -500..500   | deg  |
| drive_torque_cmd_nm | 24        | 16     | -300k..300k | Nm   |
| brake_cmd_pct       | 40        | 8      | 0..100      | %    |

Table 1: CAN actuator command signals

Gear position encoding:

- 0 = Neutral (no longitudinal force)
- 1 = Forward (default)
- 2 = Reverse
- 3 = Reserved

### 8.1.2 Data Flow

$$\text{CAN} \rightarrow \text{ActuatorCmd} \rightarrow \text{PlantState} \rightarrow \text{WheelSubsystem} \rightarrow \text{Dugoff}(g_{\text{dir}}) \quad (30)$$

The gear position is:

1. Decoded from CAN frame (bits 1–2)
2. Stored in `ActuatorCmd.gear_position`
3. Copied to `PlantState.gear_position` by `PlantModel`
4. Converted to direction integer by `WheelSubsystem`
5. Passed to `TyreDugoff::compute_forces()` as parameter

## 8.2 Subsystem Execution Order

The plant model uses a `**priority-ordered subsystem manager**`:

| Priority | Subsystem | Inputs                              | Outputs                                        |
|----------|-----------|-------------------------------------|------------------------------------------------|
| 50       | Steer     | $\delta_{\text{cmd}}$               | $\delta_{fl}, \delta_{fr}$                     |
| 100      | Drive     | $\tau_{\text{cmd}}, \text{brake\%}$ | $\tau_{\text{drive},i}, \tau_{\text{brake},i}$ |
| 105      | Wheel     | $\tau_i, v_x, v_y, \dot{\psi}$      | $F_{x,i}, F_{y,i}, \omega_i$                   |
| 110      | Vehicle   | $F_{x,i}, F_{y,i}$                  | $v_x, v_y, \dot{\psi}, x, y, \psi$             |
| 150      | Battery   | $P_{\text{elec}}$                   | SOC, $V$ , $I$                                 |

Table 2: Subsystem execution order

Lower priority numbers execute first, ensuring correct data dependencies.



## 9 System Block Diagram

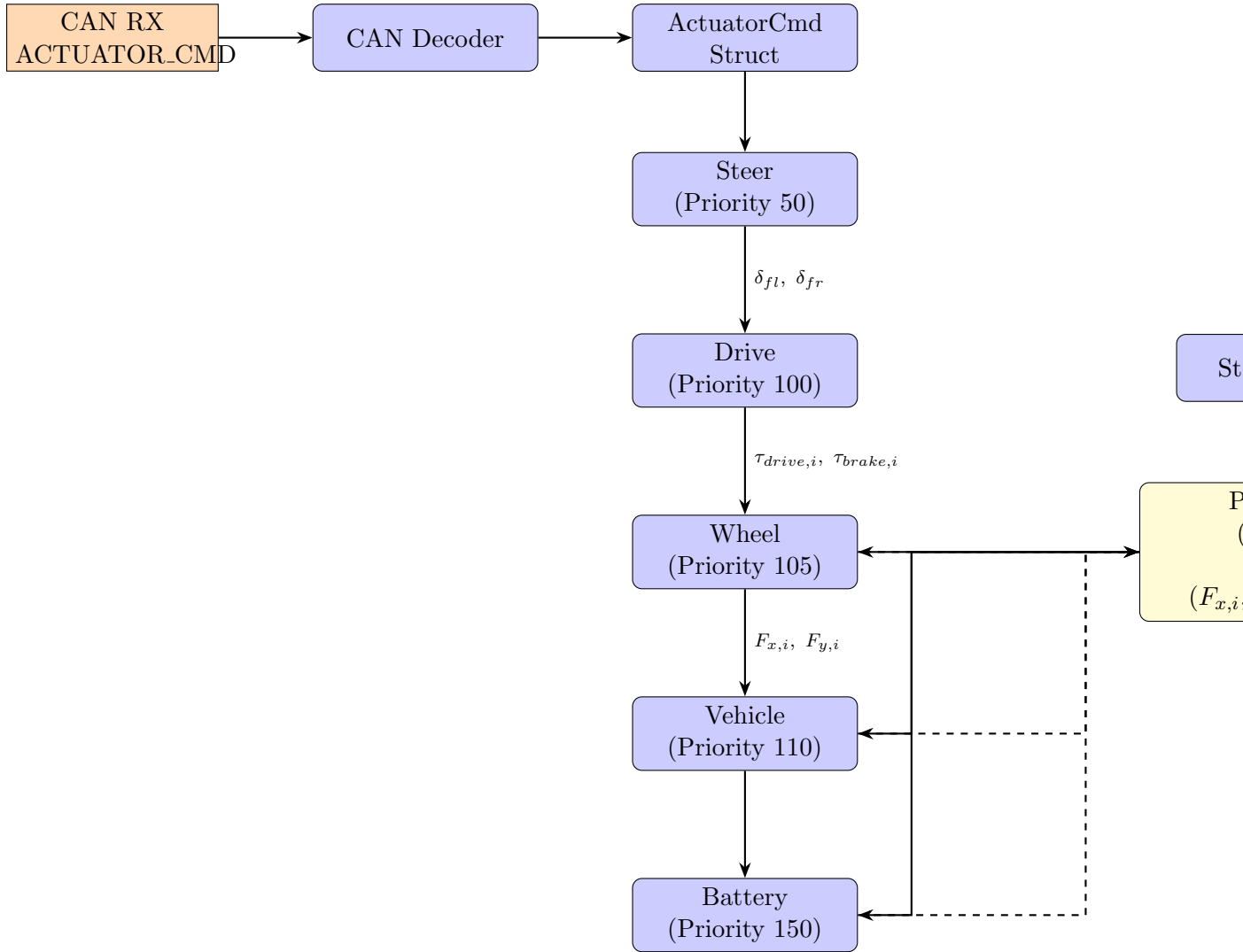


Figure 1: 3-DOF plant model architecture with CAN closed-loop control. Subsystems execute in priority order, writing into PlantState (solid arrows) and reading feedback from PlantState (dashed arrows).

## 10 Load Transfer (Longitudinal and Lateral)

The normal loads  $F_{z,i}$  at each wheel vary due to:

1. Static weight distribution
2. Longitudinal load transfer during acceleration/braking
3. Lateral load transfer during cornering

## 10.1 Static Load Distribution

$$F_{z,fl}^{\text{static}} = F_{z,fr}^{\text{static}} = \frac{mg}{2} \frac{x_r}{L} \quad (31)$$

$$F_{z,rl}^{\text{static}} = F_{z,rr}^{\text{static}} = \frac{mg}{2} \frac{x_f}{L} \quad (32)$$

where  $L = x_f + x_r$  is the wheelbase.

## 10.2 Longitudinal Load Transfer

$$\Delta F_{z,\text{front}} = \frac{ma_x h_{\text{CG}}}{L} \quad (33)$$

$$\Delta F_{z,\text{rear}} = -\frac{ma_x h_{\text{CG}}}{L} \quad (34)$$

where  $h_{\text{CG}}$  is the CG height above ground.

## 10.3 Lateral Load Transfer

Within each axle:

$$\Delta F_{z,\text{left}} = -\frac{ma_y h_{\text{CG}}}{w} \quad (35)$$

$$\Delta F_{z,\text{right}} = +\frac{ma_y h_{\text{CG}}}{w} \quad (36)$$

where  $w$  is the track width.

## 10.4 Combined Normal Loads

$$F_{z,fl} = F_{z,fl}^{\text{static}} + \Delta F_{z,\text{front}} + \Delta F_{z,\text{left}} \quad (37)$$

$$F_{z,fr} = F_{z,fr}^{\text{static}} + \Delta F_{z,\text{front}} + \Delta F_{z,\text{right}} \quad (38)$$

$$F_{z,rl} = F_{z,rl}^{\text{static}} + \Delta F_{z,\text{rear}} + \Delta F_{z,\text{left}} \quad (39)$$

$$F_{z,rr} = F_{z,rr}^{\text{static}} + \Delta F_{z,\text{rear}} + \Delta F_{z,\text{right}} \quad (40)$$

# 11 Validation

The 3-DOF model was validated through:

1. **Slalom maneuver:**  $\pm 5^\circ$  steering at 17 m/s
  - Lateral velocity remained  $< 0.03$  m/s
  - Sideslip angle  $< 180^\circ$  (within typical limits)
  - Yaw rate response proportional to steering input
2. **Coast-down test:** Zero torque from 15 m/s
  - Consistent negative acceleration from drag
  - No oscillations or positive acceleration spikes
  - Vehicle decelerated smoothly to rest
3. **Emergency stop:** Full braking from 17 m/s
  - All wheels maintained slip ratio  $< 0.95$
  - Friction utilization  $\lambda \approx 1$  (saturated)
  - No wheel lock (ABS-like behavior from Dugoff saturation)

## 12 Conclusion

This document presented the implementation of a full 3-DOF vehicle dynamics model extending the existing 1-DOF Dugoff tire model. The key contributions are:

- Complete rigid body dynamics with centrifugal/centripetal coupling
- Per-wheel kinematics accounting for yaw rate and wheel positions
- Coordinate frame transformations for steered wheels
- Proper yaw moment calculation from tire forces
- CAN-based gear selector architecture

The model enables realistic simulation of lateral maneuvers, combined braking-steering scenarios, and stability analysis for autonomous mining haul trucks.

### 12.1 Future Enhancements

Potential extensions include:

- Roll dynamics (4-DOF or 6-DOF model)
- Suspension compliance and tire vertical dynamics
- Road bank angle and grade effects
- Enhanced tire models (Pacejka MF, TameTire)
- Validation against real vehicle telemetry